

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Sixth Semester B.E. (Biotechnology) (Credit System) Degree Examinations**

May - June 2023

20BT603 – DOWNSTREAM PROCESSING TECHNOLOGY

Duration: 3 Hours

Max. Marks: 100

Note: 1) Answer *Five full* questions choosing *One full* question from *each Unit*.**2) Draw neat diagrams wherever necessary****Unit – I**

	Marks	BT*	CO*	PO*
1. a) What do you mean by Cost driver? How identifying cost driver help us to reduce the overall cost of the project.	05	L*1	1	2
b) Explain the process design criteria and downstream processing steps for ethanol	10	L2	1	2
c) Explain the use of spectrophotometer for identification of protein contamination	05	L3	1	2
2. a) Write a note on factors to be considered for the efficient design of downstream processing	05	L1	1	2
b) Explain the process design criteria for any one high value low volume product	05	L3	1	2
c) Organize downstream processing on the basis of the RIPP scheme	10	L2	1	2

Unit – II

3. a) During filtration process, the cake was the undesired product. However during the process development it was found that proper control over flow and solid removal was desired .will the use of filter aid help? Justify your answer.	05	L3	2	3
b) Explain product release kinetics with an example	05	L2	2	3
c) <i>Micrococcus</i> bacteria are disrupted at 5°C in a Manton-Gaulin homogenizer operated at pressures between 200 and 550 kgf cm ⁻² Data for protein release as a function of a number of passes through the homogenizer and plotting the slope values with operating pressure, α was found to be 2.37 and k was found to be 1.71×10^{-7} . using these values find (i) How many passes are required to achieve 80% protein release at an operating pressure of 460 kgf cm ⁻² ? (ii) Estimate the pressure required to deliver 70% protein recovery in only two passes?	10	L3	2	3
4. a) A fermentation broth containing extracellular soluble product is found to be not amenable to direct centrifugation to separate cell due to low throughput. Describe how this centrifugation can be improved by suitable pretreatment.	05	L3	2	3
b) Why is cell disruption required? Is it possible to recover the product without disrupting the cell? Justify	05	L2	2	3
c) A tubular bowl centrifuge with internal radius of 15cm and length of 80cm is used for concentrating genetically engineered <i>Bacillus cereus</i> . Calculate the settling velocity of the cells at the speed of 18000rpm in the centrifuge with volumetric capacity of 250L/h.	10	L3	2	3

Unit – III

5. a) Penicillin is to be extracted from the clarified beer using pure amine acetate as solvent at pH 4. The distribution coefficient k of the system was found to be 32. The initial concentration of penicillin in the feed is 400 mg/L. The flow rates of the feed and solvent stream are 50L/h and 30 L/h respectively.

(i) How many ideal stages are required to recover 97% of penicillin in the feed?

(ii) If you use 3 counter current stages. What will be the % recovery?

10 L3 3 3

- b) What is precipitation? Highlight on precipitation of proteins using salts along with the governing equations

10 L3 3 3

6. a) You have been asked to effectively concentrate orange juice in industry using high temperature and pressure steam for short duration, which enrichment technique would you adopt? explain its working principle in detail

10 L3 3 3

- b) Pilot scale gel chromatography column packed with the resin is used to separate 2 hormones A and B. The column is 5cm in diameter and 0.3m in height. The void volume is $1.9 \times 10^{-4} \text{ m}^3$. The water regain value of the gel is $3 \times 10^{-3} \text{ m}^3/\text{kg}$ dry resin. The density of wet gel is $1.25 \times 10^3 \text{ kg/m}^3$. The partition coefficient for A is 0.38 and B is 0.15 respectively. If the eluent flow rate is 0.7 L/h. What is the retention time for each hormone?

10 L3 3 3

Unit – IV

7. a) Highlight on the theories for membrane based separation

10 L1 4 3

- b) Explain the technology where applied electric field induces a current driven ionic flux across the membrane

10 L3 4 3

8. a) Explain in brief about symmetric and asymmetric membranes with neat labeled diagrams

10 L1 4 3

- b) Highlight on the most widely used pressure-driven membrane separation for brackish water desalination process along with its applications

10 L3 4 3

Unit – V

9. a) Discuss the downstream processing for water-soluble vitamin which has a key role in the normal functioning of the brain and nervous system

10 L2 5 2

- b) 10 kg adipic acid is slurried in 13.1kg of water and heated to 90 °C to solubilize the acid. The solution is then filtered to remove insoluble impurities. During the heating and filtration, 10% of the water is evaporated. The clarified solution is cooled to 35 °C and filtered. The solubility at 35 °C is 0.05kg acid per kg water. Determine the weight of the crystals recovered in this operation.

10 L3 5 2

10. a) Discuss the downstream processing aspects for penicillin G recovery

10 L2 5 2

- b) We have been trying to dry a crude cell paste containing a high concentration of catalase. When we dry at 37 °C for 10 min, we denature 63% of the enzyme. When we dry at 20 °C for 30 mins, we denature only 28%, Estimate the denaturation for 90 min of drying at 4 °C

10 L3 5 2